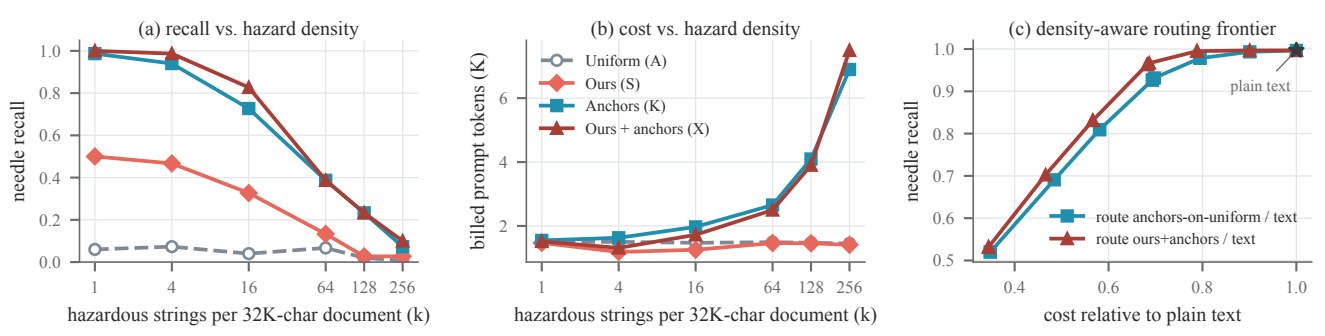
(a) recall vs. hazard density
needle recall
1.0
0.8
0.6
0.4
0.2
0.0
1
4
16
64
128
256
hazardous strings per 32K-char document (k)
(b) cost vs. hazard density
billed prompt tokens (K)
6
4
2
Uniform (A)
Ours (S)
Anchors (K)
Ours + anchors (X)
1
4
16
64
128
256
hazardous strings per 32K-char document (k)
(c) density-aware routing frontier
needle recall
1.0
0.9
0.8
0.7
0.6
0.5
plain text
route anchors-on-uniform / text
route ours+anchors / text
0.4
0.6
0.8
1.0
cost relative to plain text